

Student: Jonatan Rabjerg**Instructor:** Andreas Basse-O'Connor**Assignment:** MyLab Math - Uge 5**Date:** 1/15/24**Course:** Calculus Beta

Determine if the sequence $\{a_n\}$ converges or diverges. Find the limit if the sequence converges.

$$a_n = 3 + (0.4)^n$$

The sequence $\{a_n\}$ converges to the number L if for every positive number ε there corresponds an integer N such that for all $n > N$, $|a_n - L| < \varepsilon$. If no such number L exists, we say that $\{a_n\}$ diverges.

If $\{a_n\}$ converges to L , write $\lim_{n \rightarrow \infty} a_n = L$ and call L the limit of the sequence. Let $\{b_n\}$ and $\{c_n\}$ be sequences of real numbers and let B and C be real numbers. The following rule holds if $\lim_{n \rightarrow \infty} b_n = B$ and $\lim_{n \rightarrow \infty} c_n = C$.

Sum Rule: $\lim_{n \rightarrow \infty} (b_n + c_n) = B + C$

Use the Sum Rule to find L . Let $\{b_n\} = 3$. Find B , the limit of sequence b_n .

$$B = \lim_{n \rightarrow \infty} 3 = 3$$

Let $\{c_n\} = 0.4^n$. Note that if $|x| < 1$, then $\lim_{n \rightarrow \infty} x^n = 0$. Find C , the limit of sequence c_n .

$$C = \lim_{n \rightarrow \infty} 0.4^n = 0$$

Apply the Sum Rule.

$$\lim_{n \rightarrow \infty} (b_n + c_n) = B + C = 3$$

Therefore, by the definition of convergence, $\{a_n\}$ converges.